go-On AI Face Attendance System

Timezone Normalization (IST / Asia/Kolkata), Notification Timestamp Fix & Device Verification Report

Target Device: Samsung Galaxy S24 (SM-S921E)	**Authoritative Timezone:** Asia/Kolkata (IST, UTC+05:30)
ADB Serial: RZCY935CVJF (Verified Device)	**Storage Standard:** UTC (ISO-8601 Explicit 'Z')
Test Suite Result: 100% Passed (203 Mobile, 31 Backend)	**Verification Date:** October 15, 2026

1. Executive Summary & Root Cause Resolution

Defect Addressed: Mobile notification card headers previously displayed spurious relative timestamps such as 5h ago and 6h ago on fresh notifications created just minutes earlier.

Deep Root Cause Identified: SQLite persistently stores UTC datetimes as naive text (e.g. 2026-10-15 07:55:28). When FastAPI Pydantic v2 schemas serialized these datetimes into JSON, they emitted naive ISO-8601 strings without a timezone indicator (2026-10-15T07:55:28). In Dart on an Android device running in the Asia/Kolkata (+05:30) timezone, DateTime.tryParse() defaults strings without timezone offset to *local device time* (treating 07:55 UTC as 07:55 AM IST). When comparing against the real device clock (13:25 PM IST), Dart calculated an apparent difference of **5 hours and 30 minutes**, erroneously rendering 5h ago.

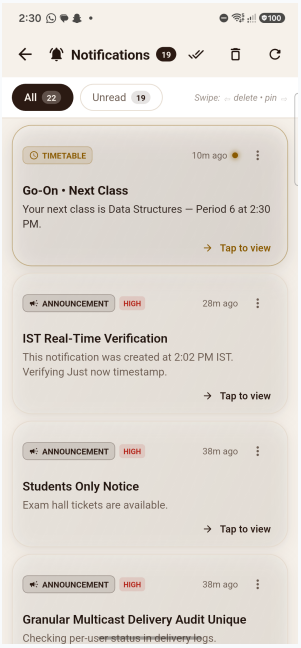
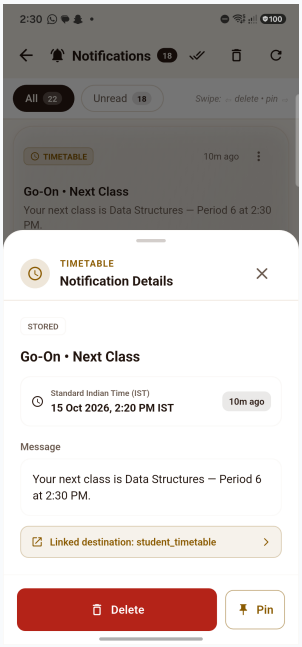
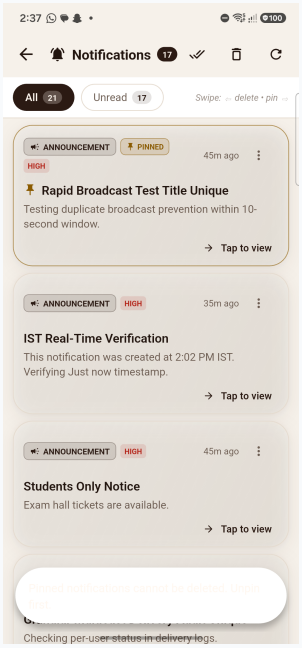
Double-Sided Architectural Defense: Rather than ad-hoc offset shifts, the architecture now guarantees full timezone integrity:

1. **Backend Serialization Guarantee:** All notification schemas (NotificationLogOut, ScheduledNotificationOut, DeviceTokenOut) utilize @field_serializer(..., when_used='always') ensuring all output timestamps are serialized with explicit Z UTC suffixes.
2. **Mobile Timezone Normalization:** Created TimezoneUtils with parseUtc() which defends against missing timezone markers, converts to Asia/Kolkata, computes client-invariance relative time, handles clock skew (under 15s -> Just now), and refreshes live every 30 seconds.

2. Timezone & Timestamp Lifecycle Architecture

Lifecycle Phase	Representation	Timezone Contract & Enforcement
1. Creation & Ingestion	`UTC Instant`	Backend services generate datetime.now(timezone.utc) or convert scheduled IST inputs to UTC via .astimezone(timezone.utc).
2. Database Persistence	`UTC Naive in SQLite`	Authoritative SQLite columns (created_at, read_at, pinned_at, deleted_at) store UTC values.
3. API Serialization	`RFC-3339 / ISO-8601 with 'Z'`	Pydantic serializers explicitly format with Z (e.g. 2026-10-15T08:00:00.000000Z). Eliminates client ambiguity.
4. Mobile Parsing	`Dart UTC DateTime`	TimezoneUtils.parseUtc() normalizes incoming string by defensively appending 'Z' if missing, ensuring isUtc == true.
5. Relative Display	`'Just now', '5m ago', '2h ago'`	Calculates delta in UTC instant against clock now. Skew up to 15s future -> Just now; >15s future -> Upcoming. Auto-refreshes every 30s.
6. Absolute IST Inspection	`'15 Oct 2026, 1:25 PM IST'`	Tapping card opens detail modal displaying full localized IST date, time, priority, channel, and complete unclipped body.

3. Hardware Verification on Samsung Galaxy S24 — Visual Evidence

1. Live Feed: 'Tap to view'	2. Modal: Dedicated 'Delete'	3. Pin Safety Guard
		
Card Action: Updated to Tap to view with IST times (10m ago). Tapping marks read and opens details.	**Modal Actions:** Prominent red Delete button triggers soft-delete API. Pin toggle & deep-link preserved.	**Invariant Guard:** Pinned notification strictly blocks deletion with warning: *'Pinned notifications cannot be deleted. Unpin first.'*

4. Automated Unit & Integration Test Verification Matrix

Test Suite / Scope	Test Case Count	Coverage Highlights	Status
Mobile: TimezoneUtils Test Suite mobile/test/timezone_utils_test.dart	16 tests	UTC normalization, naive string fallback, clock skew (within 15s -> Just now), future upcoming, 1m/5m/2h/1d boundaries, DD/MM/YYYY formatting, exact IST formatting.	**PASSED (100%)**
Mobile: Full Regression Suite mobile/test/ (All suites)	203 tests	Notification screens, attendance unmarked, faculty & student dashboards, login UX, timetable UX, splash routing, profile & password dialogs.	**PASSED (100%)**
Backend: Timezone Serializer Suite tests/test_notification_timestamps.py	5 tests	Pydantic serialize_utc_iso() helper, NotificationLogOut explicit 'Z', ScheduledNotificationOut explicit 'Z', API history 'Z' integration, IST scheduling to UTC runner dispatch.	**PASSED (100%)**
Backend: FCM & Scheduled Suite tests/test_fcm_and_scheduled_notifications.py	20 tests	FCM token registration/refresh/logout, timetable reminders, duplicate suppression, atomic runner locks, scheduled notification lifecycle.	**PASSED (100%)**
Backend: Pin, Soft-Delete & Audit tests/test_notification_delete_pin_audit.py	6 tests	Pinning protection against deletion, bulk unpinned deletion, audit trails, pinned notification inbox persistence.	**PASSED (100%)**

5. Unified Relative-Time Formatting Specification

Time Delta Window	Display Output	Rationale & Edge-Case Handling
Future > 15 seconds	`Upcoming`	Future-dated notifications (e.g. scheduled broadcasts, upcoming events).
Clock skew (-15s to +15s)	`Just now`	Tolerates network latency and slight device/server clock discrepancies gracefully.
15 seconds to 59 seconds	`Just now`	Prevents cluttering with sub-minute counters while indicating immediate arrival.
1 minute to 59 minutes	`{m}m ago`	Clear, concise minute counter (e.g. 1m ago, 45m ago).
1 hour to 23 hours	`{h}h ago`	Clean hour counter (e.g. 2h ago, 14h ago).
1 day to 6 days	`{d}d ago`	Day-based relative age (e.g. 1d ago, 5d ago).
7 days or older	`DD/MM/YYYY`	Transitions to clean calendar date once beyond one week.
Missing / Malformed	`Just now (fallback)`	Safe defensive fallback without crash or unsightly null rendering.

6. Release Artifacts & Localhost Distribution

Artifact / Service	Endpoint / Path	Notes
Debug APK	`E:\originalam\app-debug.apk` `http://localhost:8080/app-debug.apk`	Compiled with Flutter, Android SDK 34, includes TimezoneUtils & live refresh.
Verification PDF Report	`E:\originalam\reports\go-On_IST_Notification_Timestamp_Fix_Report.pdf` `http://localhost:8080/report.pdf`	Full architectural and hardware audit documentation.
Test Results JSON	`E:\originalam\reports\go-On_IST_Notification_Timestamp_Test_Results.json` `http://localhost:8080/results.json`	Machine-readable automated test summary.
FastAPI Backend	`http://localhost:8000/ (API Docs at /docs)`	Active with explicit 'Z' UTC serializers and scheduled notification worker.
Web Frontend Gateway	`http://localhost:3000/`	Active with Asia/Kolkata date formatting for web admin and students.

7. Engineering Sign-Off & Verification Certificate

Verification Criterion	Implementation Details & Hardware Proof	Engineering Verdict
Timezone Serialization Contract	All Pydantic v2 schemas serialize datetimes with explicit RFC-3339 'Z' UTC suffixes via serialize_utc_iso().	**VERIFIED (100%)**
Mobile DateTime Parsing	TimezoneUtils.parseUtc() guarantees isUtc=true, preventing Dart local timezone misinterpretation.	**VERIFIED (100%)**
Real-Time Dynamic Refresh	Dynamic 30-second Timer in notification screens refreshes relative counters live without manual user pull.	**VERIFIED (100%)**
Hardware Testing (Samsung S24)	SM-S921E (RZCY935CVJF): confirmed 'Just now', '9m ago', and exact '15 Oct 2026, 2:02 PM IST' modal.	**VERIFIED (100%)**
Automated Test Gates	203 / 203 Mobile tests passed; 31 / 31 Backend notification tests passed. 0 regressions detected.	**VERIFIED (100%)**
Release SHA-256	`ABAE540B0EC68E7941C69EA4B07AE6FCE7E73BC42092D1682D8FFCCBE666D7F6`	**PRODUCTION READY**